Zero to Hero with Git & GitHub

by Dilum De Silva

Hello, I'm

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Version Control System (VCS)

What is a version control system?

"**Version control** is a **system** that records changes to a file or set of files over time so that you can recall specific **versions** later."

- By Google 2018-10-15

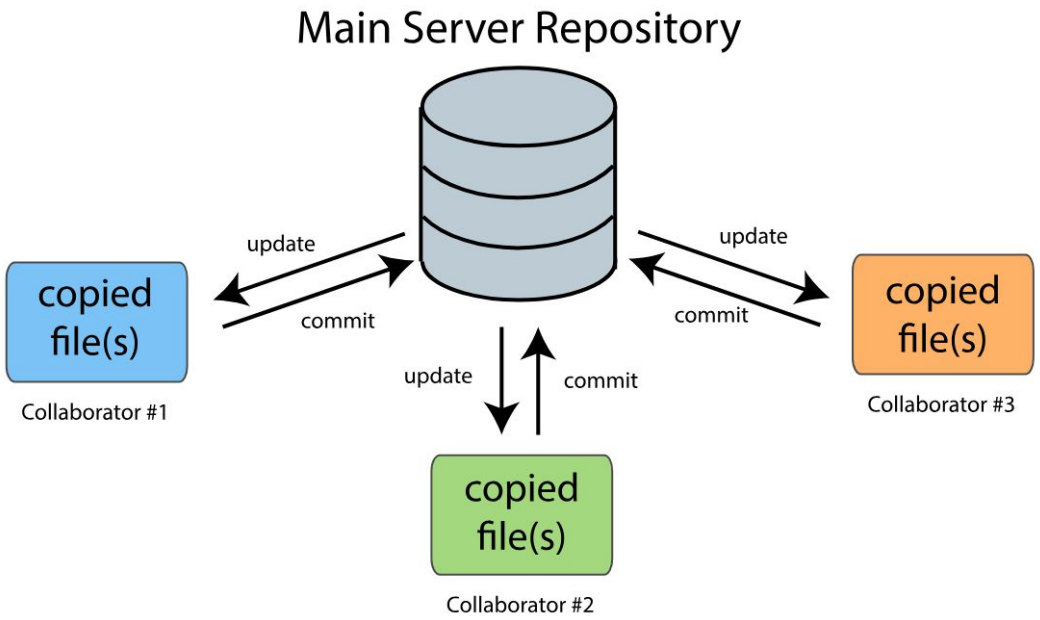
Full local history of changes made..

-by Dilum De Silva

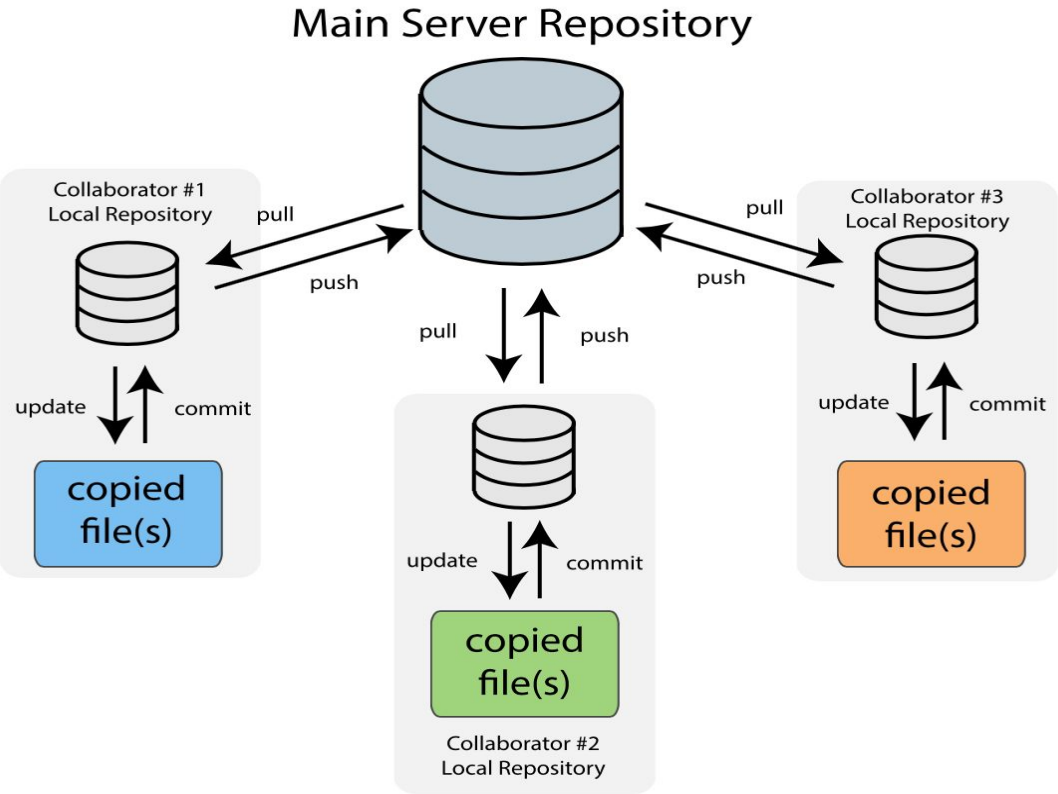
Let’s talk about types of VCS

- Local VCS
- Centralized VCS
- Distributed VCS

Centralized Version Control



Distributed Version Control



Hooray,

Now we can move with

Git & GitHub

Git is a **distributed** version control system.

It tracks our source code changes as **snapshots**

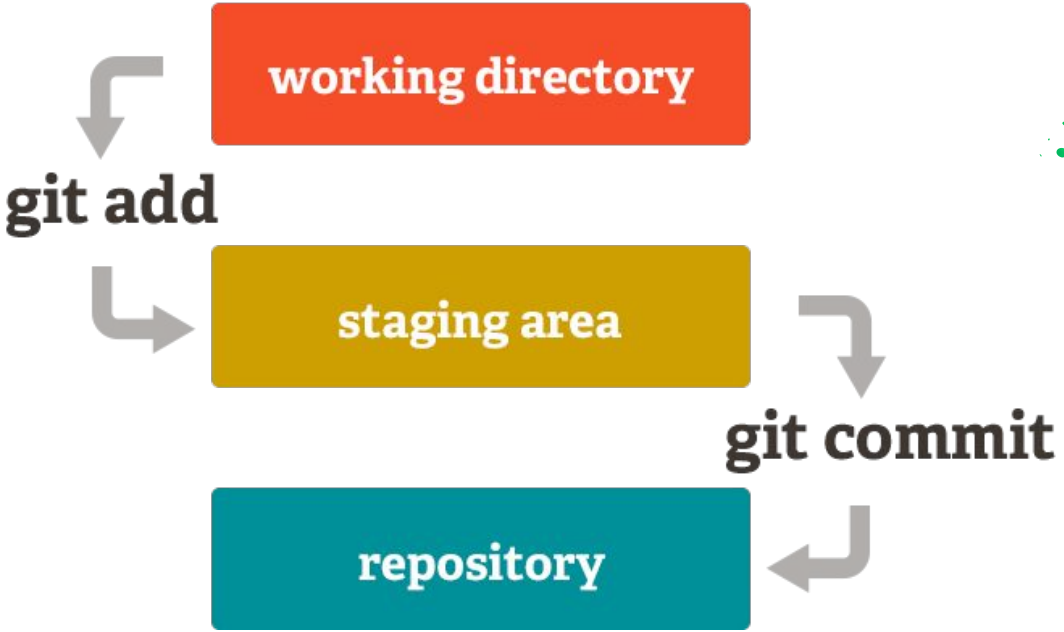
Then,
what is **GitHub ?**

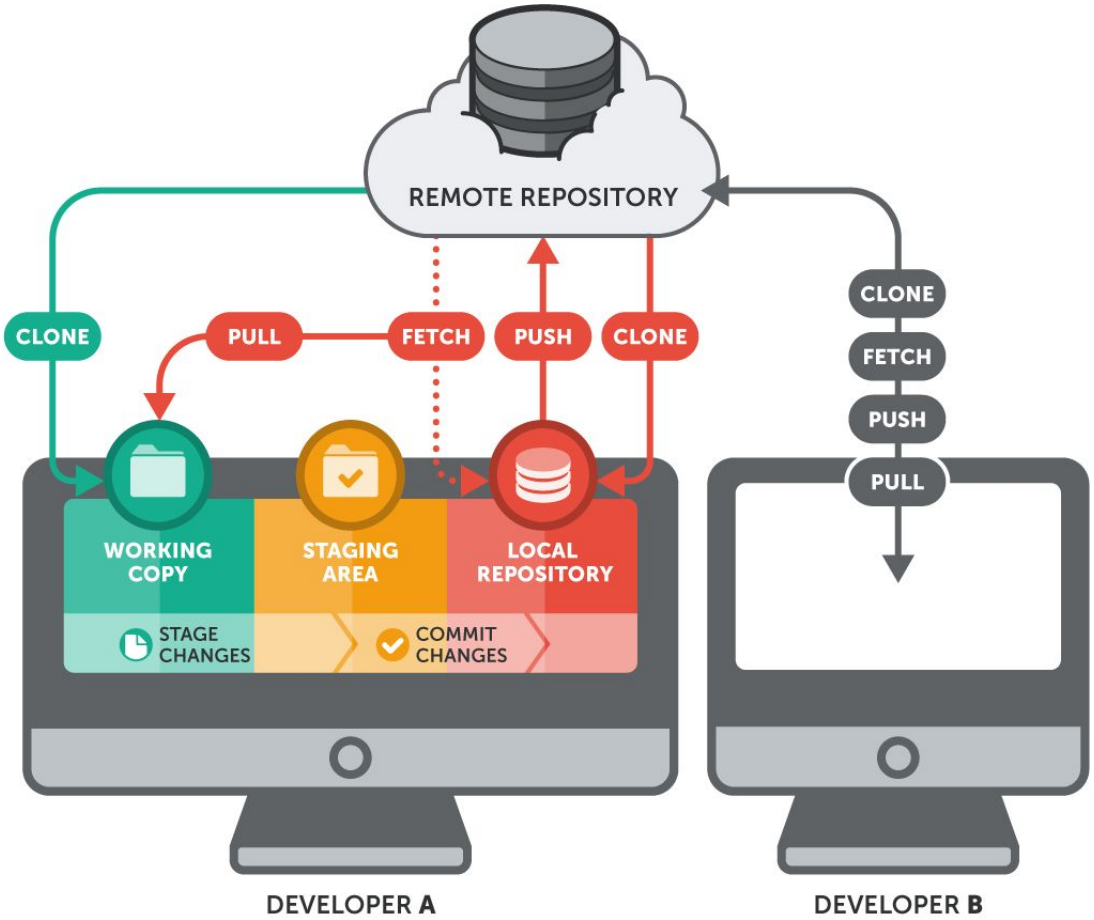
Google says,

GitHub is a web-based hosting service for version control using Git. It is mostly used for computer code.

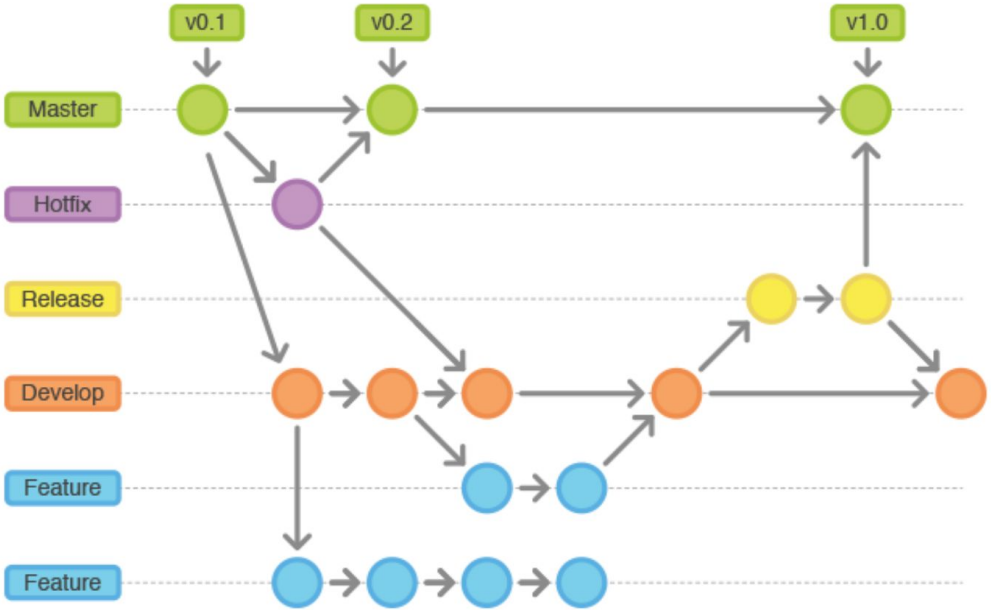
Git Workflow

Stages of git...



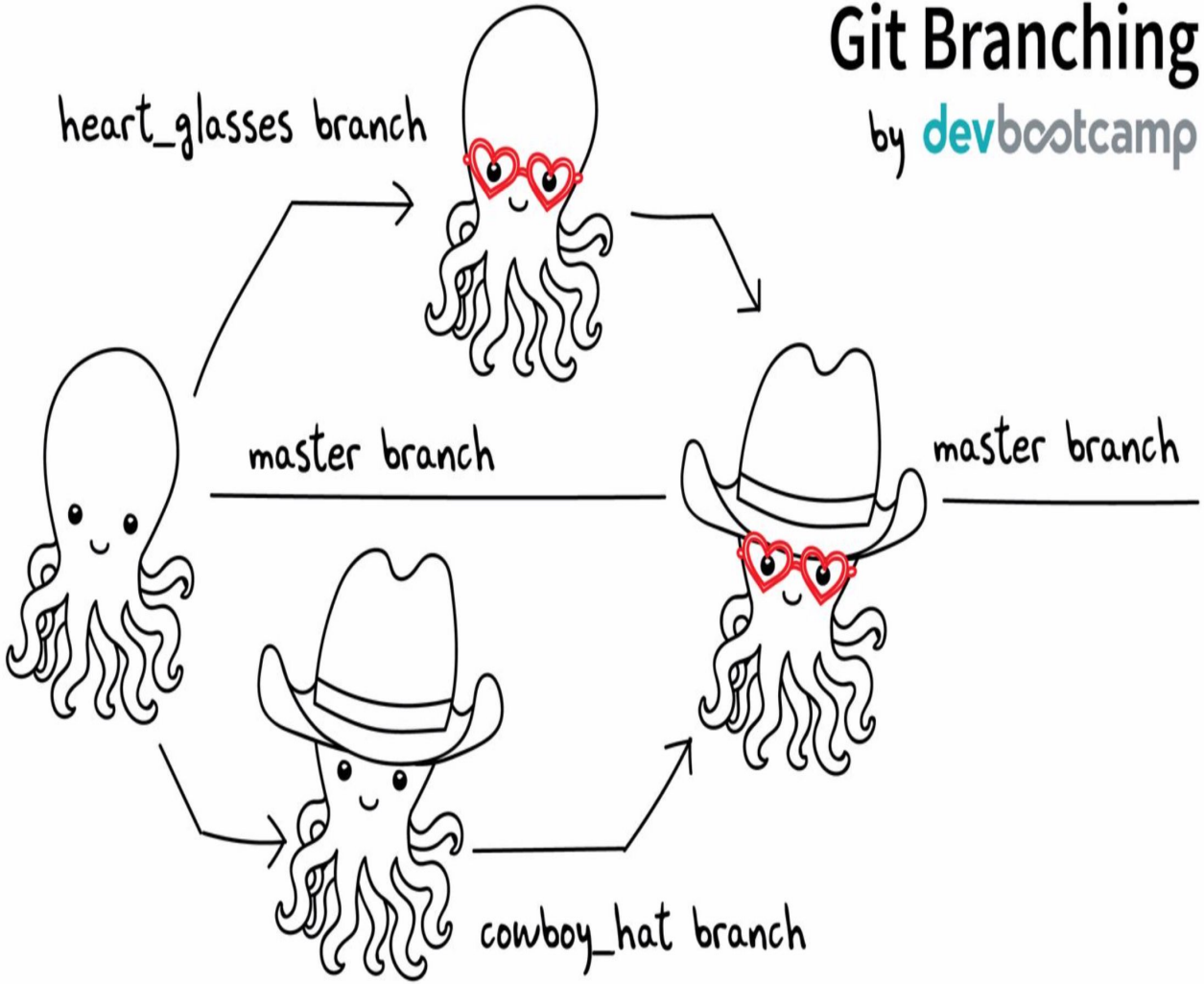

REMOTE REPOSITORY
CLONE
PULL
FETCH
PUSH
CLONE
WORKING COPY
STAGING AREA
LOCAL REPOSITORY
STAGE CHANGES
COMMIT CHANGES
DEVELOPER A
CLONE
FETCH
PUSH
PULL
DEVELOPER B

Git branch maintenance



Git Branching

by devbootcamp



Our process is **easy**

Once you use **git init** or **git clone**

It's just three steps,

git add → git commit → git push

Repos/remotes

`git init`	creates a new git repo for that folder
`git clone <url>`	create a copy of a repo in a new directory
`git remote`	lists all remote git repos
`git remote add <name> <git-url>`	adds a remote
`git rename <old-name> <new-name>`	renames a remote
`git remote remove <name>`	removes a remote
`git remote show <name>`	shows details for a remote

Branches

`git branch`	lists all local branches
`git branch -a`	lists all local ***and*** remote branches
`git branch -d <branch-name>`	deletes a branch
`git branch -m <old-name> <new-name>`	renames a branch
`git checkout <branch-name>`	switch to a different branch
`git checkout -b <new-branch-name>`	create a new branch & switch to it

File changes

`git status`	show differences between index and current HEAD commit
`git diff`	show changes between commits
`git reflog`	lists all of the commit history
`git log`	detailed version of the commit history

Commits

`git add <name>`	stage new file for new commit
`git commit -m <message>`	add snapshot of code
`git reset [<type>] [<commit>]`	unstage file & discard uncommitted changes
`git revert [<type>] [<commit>]`	undo commits

Types:

--soft : keeps changes made since last commit & resets current branch to **<commit>**

--hard : discards changes made since last commit & resets current branch to **<commit>**

Merging objects

`git push <remote> <branch>`	adds objects to merge the diffs in the remote git repo
`git pull <remote> <branch>`	gets objects from the remote repo and merges the diffs
`git fetch [<remote>]`	gets objects from remote repo, but doesn't merge in the diffs

Demo Time

Thanks!

Any questions?

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”Stay hungry, stay foolish”
-Steve Jobs